

Marc Najork

EXPERIENCE

Mar 2014–now	Google Inc., Mountain View, CA <i>Distinguished Research Scientist</i> (April 2023–now) <i>Senior Director, Research Engineering</i> (Nov 2021–April 2023) <i>Director, Research Engineering</i> (Nov 2017–Oct 2021) <i>Senior Staff Research Scientist</i> (Mar 2014–Oct 2017) • Since May 2023, individual contributor in Google DeepMind. Exploring challenges in Generative Information Retrieval. • From Jan. 2015 until April 2023, managed a group of (at peak) 69 FTE in Google Research. Virtually all of our research output is used in Google products. Cumulative revenue impact ~\$5B ARR. • From Mar. 2014 until Dec. 2014, part of the Personal Search Infrastructure team. Worked on HappyHour, a processing and serving system for structured personal data.
Oct 2001–Mar 2014	Microsoft Corporation, Microsoft Research Silicon Valley, Mountain View, CA <i>Principal Researcher</i> (Mar 2006–Mar 2014) <i>Senior Researcher</i> (Oct 2005–Mar 2006) <i>Researcher</i> (Oct 2001–Sep 2005) • Collaborated with Bing on various facets of social search. • Explored link-based ranking techniques for web search results. • Developed the Scalable Hyperlink Store, a specialized database giving extremely fast access to nodes and edges of the web graph induced by the Bing corpus. • Explored techniques for identifying spam web pages. • Consulted on large-scale web crawling for Bing. • Principal contributor to PageTurner, a large-scale study of the evolution of web pages. • Contributed to Boxwood, a distributed, scalable, and reliable B-Tree system. • Four Microsoft Gold Star awards; three Microsoft Research Tech Transfer Awards; Microsoft Corporate Bench Program class of 2005; promoted to Partner level in 2009.
Oct 1993–Sep 2001	Digital Equipment Corporation (Compaq since 1998), Systems Research Center, Palo Alto, CA <i>Manager, Programming Technology</i> (Jun 2001–Sep 2001) <i>Senior Member of Technical Staff</i> (Sep 1999–Jun 2001) <i>Software Principal Engineer</i> (Oct 1993–Sep 1999) • Managed a group of five Ph.D.-level researchers. Responsible for four projects. • Principal contributor to Mercator, an extensible, high-performance web crawler. Mercator formed the web crawling component of AltaVista's Search Engine 3 product, which sold to over 1,200 enterprise customers, and it became the standard web crawler of the various AltaVista sites. • Main contributor to JCAT, a Java-based algorithm animation system. • Worked on tools and techniques for building distributed, collaborative, web-based applications. • Designed and implemented Obliq-3D, a fast-turnaround, interactive 3D animation environment.

EDUCATION

January 1994	Ph.D. in Computer Science, University of Illinois at Urbana-Champaign Dissertation " <i>Programming in Three Dimensions</i> " supervised by Prof. Simon Kaplan.
May 1989	Diplom-Wirtschaftsinformatiker, Technical University of Darmstadt, Germany Program covers Computer Science, Mathematics, Business Administration, Economics, and Law.

HONORS

2025	Fellow of the Web Intelligence Academy (WIA)
2023	Fellow of the American Association for the Advancement of Science (AAAS) " <i>For distinguished contributions to the field of information retrieval, specifically to web search and web content analysis</i> "
2021	Elected member of the ACM SIGIR Academy
2020	Fellow of the Institute of Electrical and Electronics Engineers (IEEE) " <i>For contributions to web crawling and web data processing</i> "
2019	Fellow of the Association for Computing Machinery (ACM) " <i>For contributions to web search and web science</i> "

AWARDS

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- 2026 SIGIR Test of Time Award
for *Learning to Rank with Selection Bias in Personal Search* (SIGIR 2016)
- 2019 Best Short Paper Award
for *An Analysis of the Softmax Cross Entropy Loss for Learning-to-Rank with Binary Relevance* (ICTIR 2019)
- 2017 Best Paper Honorable Mention
for *Learning from User Interactions in Personal Search via Attribute Parameterization* (WSDM 2017)
- 1996 Best Paper Award
for *Collaborative Active Textbooks: A Web-Based Algorithm Animation System for an Electronic Classroom* (VL 1996)

PATENTS

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- P45 | US Patent 12,718,022. *Systems and methods for using document document activity logs to train machine-learned models for determining document relevance*. Inventors: Weize Kong, Michael Bendersky, **Marc Najork**, Rama Kumar Pasumarthi, Zhen Qin, Rolf Jagerman. Assignee: Google LLC. Filed 2020-06-15, issued 2026-08-25.
- P44 | US Patent 12,585,712. *Adversarial bandits policy for crawling highly dynamic content*. Inventors: Michael Bendersky, Przemyslaw Gajda, Sergey Novikov, **Marc Alexander Najork**, Shuguang Han. Assignee: Google LLC. Filed 2024-11-04, issued 2026-03-24.
- P43 | US Patent 12,393,768. *Layout-aware multimodal pretraining for multimodal document understanding*. Inventors: Mingyang Zhang, Cheng Li, Tao Chen, Spurthi Amba Hombaiah, Michael Bendersky, **Marc Alexander Najork**, Te-Lin Wu. Assignee: Google LLC. Filed 2020-12-22, issued 2025-08-19.
- P42 | US Patent 12,354,396. *System for information extraction from form-like documents*. Inventors: Sandeep Tata, Bodhisattwa Majumder, Qi Zhao, James Bradley Wendt, **Marc Najork**, Navneet Potti. Assignee: Google LLC. Filed 2023-10-19, issued 2025-07-08.
- P41 | US Patent 12,236,322. *Training and/or utilizing a model for predicting measures reflecting both quality and popularity of content*. Inventors: Spurthi Amba Hombaiah, Vladimir Ofitserov, Mike Bendersky, **Marc Alexander Najork**. Assignee: Google LLC. Filed 2022-12-05, issued 2025-02-25.
- P40 | US Patent 12,210,837. *Systems and Methods for Machine-Learned Prediction of Semantic Similarity Between Documents*. Inventors: Liu Yang, **Marc Najork**, Michael Bendersky, Mingyang Zhang, Cheng Li. Assignee: Google LLC. Filed 2023-05-22, issued 2025-01-28.
- P39 | US Patent 12,182,509. *Processing large-scale textual inputs using neural networks*. Inventors: Karthik Raman, Liu Yang, Mike Bendersky, Jiecao Chen, **Marc Alexander Najork**. Assignee: Google LLC. Filed 2021-06-01, issued 2024-12-31.
- P38 | US Patent 12,141,214. *Adversarial bandits policy for crawling highly dynamic content*. Inventors: Michael Bendersky, Przemyslaw Gajda, Sergey Novikov, **Marc Alexander Najork**, Shuguang Han. Assignee: Google LLC. Filed 2020-03-30, issued 2024-11-12.
- P37 | US Patent 12,072,839. *Automatic file organization within a cloud storage system*. Inventors: Weize Kong, Mingyang Zhang, Michael Bendersky, **Marc Alexander Najork**, Mike Colagrosso, Brandon Vargo, Remy Burger. Assignee: Google LLC. Filed 2021-12-07, issued 2024-08-27.
- P36 | US Patent 11,830,269. *System for information extraction from form-like documents*. Inventors: Sandeep Tata, Bodhisattwa Majumder, Qi Zhao, James Bradley Wendt, **Marc Najork**, Navneet Potti. Assignee: Google LLC. Filed 2022-07-18, issued 2023-11-28.
- P35 | US Patent 11,694,034. *Systems and methods for machine-learned prediction of semantic similarity between documents*. Inventors: Liu Yang, **Marc Najork**, Michael Bendersky, Mingyang Zhang, Cheng Li. Assignee: Google LLC. Filed 2020-10-23, issued 2023-07-04.
- P34 | US Patent 11,551,150. *Training and/or utilizing a model for predicting measures reflecting both quality and popularity of content*. Inventors: Spurthi Amba Hombaiah, Vladimir Ofitserov, Mike Bendersky, **Marc Alexander Najork**. Assignee: Google LLC. Filed 2020-07-06, issued 2023-01-10.
- P33 | US Patent 11,526,752. *Systems and methods for active learning*. Inventors: Qi Zhao, Abbas Kazerouni, Sandeep Tata, Jing Xie, **Marc Najork**. Assignee: Google LLC. Filed 2020-01-23, issued 2022-12-13.

- P32 | US Patent 11,393,233. *System for information extraction from form-like documents*. Inventors: Sandeep Tata, Bodhisattwa Majumder, Qi Zhao, James Bradley Wendt, **Marc Najork**, Navneet Potti. Assignee: Google LLC. Filed 2020-06-02, issued 2022-07-19.
- P31 | US Patent 11,238,058. *Search and retrieval of structured information cards*. Inventors: **Marc Alexander Najork**, Sujith Ravi, Michael Bendersky, Peter Shao-sen Young, Timothy Youngjin Sohn, Mingyang Zhang, Thomas Nelson, Xuanhui Wang. Assignee: Google LLC. Filed 2020-11-02, issued 2022-02-01.
- P30 | US Patent 10,970,293. *Ranking search result documents*. Inventors: Mike Bendersky, **Marc Alexander Najork**, Donald Metzler, Xuanhui Wang. Assignee: Google LLC. Filed 2019-08-26, issued 2021-04-06.
- P29 | US Patent 10,824,630. *Search and retrieval of structured information cards*. Inventors: **Marc Alexander Najork**, Sujith Ravi, Michael Bendersky, Peter Shao-sen Young, Timothy Youngjin Sohn, Mingyang Zhang, Thomas Nelson, Xuanhui Wang. Assignee: Google LLC. Filed 2016-10-26, issued 2020-11-03.
- P28 | US Patent 10,540,610. *Generating and applying a trained structured machine learning model for determining a semantic label for content of a transient segment of a communication*. Inventors: Jie Yang, Amr Ahmed, Luis Garcia Pueyo, Mike Bendersky, Amitabh Saikia, Marc-Allen Cartright, **Marc Alexander Najork**, MyLinh Yang, Hui Tan, Weinan Zhang, Vanja Josifovski, Alexander J. Smola. Assignee: Google LLC. Filed 2016-04-27, issued 2020-01-21.
- P27 | US Patent 10,394,832. *Ranking search result documents*. Inventors: Mike Bendersky, **Marc Alexander Najork**, Donald Metzler, Xuanhui Wang. Assignee: Google LLC. Filed 2016-10-24, issued 2019-08-27.
- P26 | US Patent 9,953,185. *Identifying query patterns and associated aggregate statistics among search queries*. Inventors: Mike Bendersky, Donald Metzler, **Marc Alexander Najork**, Dor Naveh, Vlad Panait, Xuanhui Wang. Assignee: Google LLC. Filed 2015-11-24, issued 2017-05-25.
- P25 | US Patent 8,949,232. *Social network recommended content and recommending members for personalized search results*. Inventors: Timothy Harrington, Rajesh Shenoy, **Marc Najork**, Rina Panigrahy. Assignee: Microsoft Corporation. Filed 2011-10-04, issued 2015-02-03.
- P24 | US Patent 8,856,112. *Considering document endorsements when processing queries*. Inventors: **Marc A. Najork**, Rina Panigrahy, Rajesh K. Shenoy. Assignee: Microsoft Corporation. Filed 2011-08-26, issued 2014-10-01.
- P23 | US Patent 8,666,920. *Estimating shortest distances in graphs*. Inventors: **Marc A. Najork**, Sreenivas Gollapudi, Rina Panigrahy, Atish Das Sarma. Assignee: Microsoft Corporation. Filed 2011-08-18, issued 2014-03-04.
- P22 | US Patent 8,392,366. *Changing number of machines running distributed hyperlink database*. Inventors: **Marc A. Najork**. Assignee: Microsoft Corporation. Filed 2006-08-29, issued 2013-03-05.
- P21 | US Patent 8,209,305. *Incremental update scheme for hyperlink database*. Inventors: **Marc A. Najork**. Assignee: Microsoft Corporation. Filed 2007-10-25, issued 2012-06-26.
- P20 | US Patent 7,962,510. *Using content analysis to detect spam web pages*. Inventors: **Marc Alexander Najork**, Dennis Craig Fetterly, Mark Steven Manasse, Alexandros Ntoulas. Assignee: Microsoft Corporation. Filed 2005-02-11, issued 2011-06-14.
- P19 | US Patent 7,818,334. *Query dependent link-based ranking using authority scores*. Inventors: **Marc A. Najork**. Assignee: Microsoft Corporation. Filed 2007-10-22, issued 2010-10-19.
- P18 | US Patent 7,792,854. *Query dependent link-based ranking*. Inventors: **Marc A. Najork**. Assignee: Microsoft Corporation. Filed 2007-10-22, issued 2010-09-07.
- P17 | US Patent 7,783,671. *Deletion and compaction using versioned nodes*. Inventors: **Marc A. Najork**, Chandramohan A. Thekkath. Filed 2006-03-16, issued 2010-08-24.
- P16 | US Patent 7,739,281. *Systems and methods for ranking documents based upon structurally interrelated information*. Inventors: **Marc A. Najork**. Assignee: Microsoft Corporation. Filed 2003-09-16, issued 2010-06-15.
- P15 | US Patent 7,680,785. *Systems and methods for inferring uniform resource locator (URL) normalization rules*. Inventors: **Marc Alexander Najork**. Assignee: Microsoft Corporation. Filed 2005-03-25, issued 2010-03-16.

- P14 | US Patent 7,627,777. *Fault tolerance scheme for distributed hyperlink database*. Inventors: **Marc Alexander Najork**. Assignee: Microsoft Corporation. Filed 2006-03-17, issued 2009-12-01.
- P13 | US Patent 7,340,467. *System and method for maintaining a distributed database of hyperlinks*. Inventors: **Marc A. Najork**. Assignee: Microsoft Corporation. Filed 2003-04-15, issued 2008-03-04.
- P12 | US Patent 7,139,747. *System and method for distributed web crawling*. Inventors: **Marc Alexander Najork**. Assignee: Hewlett-Packard Development Company. Filed 2000-11-03, issued 2006-11-21.
- P11 | US Patent 7,082,438. *Algorithm for tree traversals using left links*. Inventors: **Marc A. Najork**, Chandramohan A. Thekkath. Assignee: Microsoft Corporation. Filed 2005-09-01, issued 2006-07-25.
- P10 | US Patent 7,072,904. *Deletion and compaction using versioned nodes*. Inventors: **Marc A. Najork**, Chandramohan A. Thekkath. Assignee: Microsoft Corporation. Filed 2002-12-02, issued 2006-07-04.
- P9 | US Patent 7,007,027. *Algorithm for tree traversals using left links*. Inventors: **Marc A. Najork**, Chandramohan A. Thekkath. Assignee: Microsoft Corporation. Filed 2002-12-02, issued 2006-02-28.
- P8 | US Patent 6,952,730. *System and method for efficient filtering of data set addresses in a web crawler*. Inventors: **Marc Alexander Najork**, Clark Allan Heydon. Assignee: Hewlett-Packard Development Company. Filed 2000-06-30, issued 2005-10-04.
- P7 | US Patent 6,910,077. *System and method for identifying cloaked web servers*. Inventors: **Marc A. Najork**. Assignee: Hewlett-Packard Development Company. Filed 2002-01-04, issued 2005-06-21.
- P6 | US Patent 6,594,694. *System and method for near-uniform sampling of web page addresses*. Inventors: **Marc Alexander Najork**, Clark Allan Heydon, Michael Mitzenmacher, Monika H. Henzinger. Assignee: Hewlett-Packard Development Company. Filed 2000-05-12, issued 2003-07-15.
- P5 | US Patent 6,377,984. *Web crawler system using parallel queues for queuing data sets having common address and concurrently downloading data associated with data set in each queue*. Inventors: **Marc Alexander Najork**, Clark Allan Heydon. Assignee: Alta Vista Company. Filed 1999-11-02, issued 2002-04-23.
- P4 | US Patent 6,351,755. *System and method for associating an extensible set of data with documents downloaded by a web crawler*. Inventors: **Marc Alexander Najork**, Clark Allan Heydon. Assignee: Alta Vista Company. Filed 1999-11-02, issued 2002-02-26.
- P3 | US Patent 6,321,265. *System and method for enforcing politeness while scheduling downloads in a web crawler*. Inventors: **Marc Alexander Najork**, Clark Allan Heydon. Assignee: Alta Vista Company. Filed 1999-11-02, issued 2001-11-20.
- P2 | US Patent 6,301,614. *System and method for efficient representation of data set addresses in a web crawler*. Inventors: **Marc Alexander Najork**, Clark Allan Heydon. Assignee: Alta Vista Company. Filed 1999-11-02, issued 2001-10-09.
- P1 | US Patent 6,263,364. *Web crawler system using plurality of parallel priority level queues having distinct associated download priority levels for prioritizing document downloading and maintaining document freshness*. Inventors: **Marc Alexander Najork**, Clark Allan Heydon, Janet Lynn Wiener. Assignee: Alta Vista Company. Filed 1999-11-02, issued 2001-07-17.

For a list of non-issued published patent applications please visit my home page
<http://marc.najork.org>

CONFERENCE AND WORKSHOP PAPERS

- C100 | Rongzhi Zhang, Jiaming Shen, Tianqi Liu, Jialu Liu, Michael Bendersky, **Marc Najork**, Chao Zhang. *Knowledge Distillation with Perturbed Loss: From a Vanilla Teacher to a Proxy Teacher*. In 2024 ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pages 4278–4289, August 2024.
- C99 | Spurthi Amba Hombaiah, Tao Chen, Mingyang Zhang, Michael Bendersky, **Marc Najork**, Matt Colen, Sergey Levi, Vladimir Ofitserov, Tanvir Amin. *Creator Context for Tweet Recommendation*. In Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing: Industry Track, pages 353–363, December 2023.

- C98 | Sanket Vaibhav Mehta, Jai Gupta, Yi Tay, Mostafa Dehghani, Vinh Q. Tran, Jinfeng Rao, **Marc Najork**, Emma Strubell, Donald Metzler. *DSI++: Updating Transformer Memory with New Documents*. In Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, pages 8198–8213, December 2023.
- C97 | Aijun Bai, Rolf Jagerman, Zhen Qin, Le Yan, Pratyush Kar, Bing-Rong Lin, Xuanhui Wang, Michael Bendersky, **Marc Najork**. *Regression Compatible Listwise Objectives for Calibrated Ranking*. In Proceedings of the 32nd ACM International Conference on Information and Knowledge Management, pages 4502–4508, October 2023.
- C96 | Yunan Zhang, Le Yan, Zhen Qin, Honglei Zhuang, Jiaming Shen, Xuanhui Wang, Michael Bendersky, **Marc Najork**. *Towards Disentangling Relevance and Bias in Unbiased Learning to Rank*. In Proceedings of the 2023 ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pages 5618–5627, August 2023.
- C95 | Karan Samel, Cheng Li, Weize Kong, Tao Chen, Mingyang Zhang, Shaleen Gupta, Swaraj Khadanga, Wensong Xu, Xingyu Wang, Kashyap Kolipaka, Michael Bendersky, **Marc Najork**. *End-to-End Query Term Weighting*. In Proceedings of the 2023 ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pages 4778–4786, August 2023.
- C94 | Aditi Chaudhary, Karthik Raman, Krishna Srinivisan, Kazuma Hashimoto, Mike Bendersky, **Marc Najork**. *Exploring the Viability of Synthetic Query Generation for Relevance Prediction*. In Proceedings of the SIGIR 2023 Workshop on eCommece, July 2023.
- C93 | Jiaming Shen, Jialu Liu, Dan Finnie, Negar Rahmati, Michael Bendersky, **Marc Najork**. *“Why is this misleading?” Detecting News Headline Hallucinations with Explanations*. In Proceedings of the ACM Web Conference 2023, pages 1662–1672, April 2023.
- C92 | Cheng Li, Yaping Qi, Hayk Zakaryan, Mingyang Zhang, Michael Bendersky, Yonghua Wu, **Marc Najork**. *Job Type Extraction for Service Businesses*. In Companion Proceedings of the ACM Web Conference 2023, pages 401–405, April 2023.
- C91 | Beliz Gunel, Sandeep Tata, **Marc Najork**. *STRUM: Extractive Aspect-Based Contrastive Summarization*. In Companion Proceedings of the ACM Web Conference 2023, pages 28–31, April 2023.
- C90 | Rolf Jagerman, Xuanhui Wang, Honglei Zhuang, Zhen Qin, Michael Bendersky, **Marc Najork**. *Rax: Composable Learning-to-Rank using JAX*. In Proceedings of the 28th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pages 3051–3060, August 2022.
- C89 | Le Yan, Zhen Qin, Xuanhui Wang, Michael Bendersky, **Marc Najork**. *Scale Calibration of Deep Ranking Models*. In Proceedings of the 28th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pages 4300–4309, August 2022.
- C88 | Rolf Jagerman, Zhen Qin, Xuanhui Wang, Michael Bendersky, **Marc Najork**. *On Optimizing Top-K Metrics for Neural Ranking Models*. In Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval, pages 2303–2307, July 2022.
- C87 | Le Yan, Zhen Qin, Honglei Zhuang, Xuanhui Wang, Michael Bendersky, **Marc Najork**. *Revisiting Two-tower Models for Unbiased Learning to Rank*. In Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval, pages 2410–2414, July 2022.
- C86 | Tao Chen, Mingyang Zhang, Jing Lu, Michael Bendersky, **Marc Najork**. *Out-of-Domain Semantics to the Rescue! Zero-Shot Hybrid Retrieval Models*. In Proceedings of the 44th European Conference on Information Retrieval, pages 95–110, April 2022.
- C85 | Chen Qu, Weize Kong, Liu Yang, Mingyang Zhang, Michael Bendersky, **Marc Najork**. *Natural Language Understanding with Privacy-Preserving BERT*. In Proceedings of the 30th ACM International Conference on Information and Knowledge Management, pages 1488–1497, November 2021.
- C84 | Spurthi Amba Hombaiah, Tao Chen, Mingyang Zhang, Michael Bendersky, **Marc Najork**. *Dynamic Language Models for Continuously Evolving Content*. In Proceedings of the 27th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pages 2514–2524, August 2021.
- C83 | Zhen Qin, Honglei Zhuang, Rolf Jagerman, Xinyu Qian, Po Hu, Dan Chary Chen, Xuanhui Wang, Michael Bendersky, **Marc Najork**. *Bootstrapping Recommendations at Chrome Web Store*. In Proceedings of the 27th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pages 3483–3491, August 2021.

- C82 | Nicholas Monath, Kumar Avinava Dubey, Guru Guruganesh, Manzil Zaheer, Amr Ahmed, Andrew McCallum, Gokhan Mergen, **Marc Najork**, Mert Terzihan, Bryon Tjanaka, Yuan Wang, Yuchen Wu. *Scalable Agglomerative Clustering*. In Proceedings of the 27th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pages 1245–1255, August 2021.
- C81 | Beliz Gunel, Navneet Potti, Sandeep Tata, James Wendt, **Marc Najork**, Jing Xie. *Data-Efficient Information Extraction from Form-Like Documents*. In Proceedings of the 2nd Document Intelligence Workshop at KDD 2021, August 2021.
- C80 | Krishna Srinivasan, Karthik Raman, Jiecao Chen, Michael Bendersky, **Marc Najork**. *WIT: Wikipedia-based Image Text Dataset for Multimodal Multilingual Machine Learning*. In Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval, pages 2443–2449, July 2021.
- C79 | Honglei Zhuang, Zhen Qin, Shuguang Han, Xuanhui Wang, Michael Bendersky, **Marc Najork**. *Ensemble Distillation for BERT-Based Ranking Models*. In Proceedings of the 7th ACM SIGIR International Conference on the Theory of Information Retrieval, pages 131–136, July 2021.
- C78 | Zhen Qin, Le Yan, Honglei Zhuang, Yi Tay, Rama Kumar Pasumarthi, Xuanhui Wang, Michael Bendersky, **Marc Najork**. *Are Neural Raters Still Outperformed by Gradient Boosted Decision Trees?*. In Proceedings of the 9th International Conference on Learning Representations, May 2021.
- C77 | Rolf Jagerman, Weize Kong, Rama Kumar Pasumarthi, Zhen Qin, Michael Bendersky, **Marc Najork**. *Improving Cloud Storage Search with User Activity*. In Proceedings of the 14th ACM International Conference on Web Search and Data Mining, pages 508–516, March 2021.
- C76 | Jiecao Chen, Liu Yang, Karthik Raman, Michael Bendersky, Jung-Jung Yeh, Yun Zhou, **Marc Najork**, Danyang Cai, Ehsan Emadzadeh. *DiPair: Fast and Accurate Distillation for Trillion-Scale Text Matching and Pair Modeling*. In Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: Findings, pages 2925–2937, November 2020.
- C75 | Liu Yang, Mingyang Zhang, Cheng Li, Michael Bendersky, **Marc Najork**. *Beyond 512 Tokens: Siamese Multi-depth Transformer-based Hierarchical Encoder for Long-Form Document Matching*. In Proceedings of the 29th ACM International Conference on Information and Knowledge Management, pages 1725–1734, October 2020.
- C74 | Rama Kumar Pasumarthi, Honglei Zhuang, Xuanhui Wang, Michael Bendersky, **Marc Najork**. *Permutation Equivariant Document Interaction Network for Neural Learning to Rank*. In Proceedings of the 6th ACM SIGIR International Conference on the Theory of Information Retrieval, pages 145–148, September 2020.
- C73 | Weize Kong, Michael Bendersky, **Marc Najork**, Brandon Vargo, Mike Colagrosso. *Learning to Cluster Documents into Workspaces Using Large Scale Activity Logs*. In Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pages 2416–2424, August 2020.
- C72 | Honglei Zhuang, Xuanhui Wang, Michael Bendersky, **Marc Najork**. *Feature Transformation for Neural Ranking Models*. In Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval, pages 1649–1652, July 2020.
- C71 | Bodhisattwa Prasad Majumder, Navneet Potti, Sandeep Tata, James B. Wendt, Qi Zhao, **Marc Najork**. *Representation Learning for Information Extraction from Form-like Documents*. In Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, pages 6495–6504, July 2020.
- C70 | Shuguang Han, Michael Bendersky, Przemek Gajda, Sergey Novikov, **Marc Najork**, Bernhard Brodowsky, Alexandrin Popescul. *Adversarial Bandits Policy for Crawling Commercial Web Content*. In Proceedings of the 2020 World Wide Web Conference, pages 407–417, April 2020.
- C69 | Sebastian Bruch, Shuguang Han, Michael Bendersky, **Marc Najork**. *A Stochastic Treatment of Learning to Rank Scoring Functions*. In Proceedings of the 13th ACM International Conference on Web Search and Data Mining, pages 61–69, February 2020.
- C68 | Ying Sheng, Nguyen Vo, James B. Wendt, Sandeep Tata, **Marc Najork**. *Migrating a Privacy-Safe Information Extraction System to a Software 2.0 Design*. In Proceedings of the 10th Conference on Innovative Data Systems Research, January 2020.

- C67 | Qingyao Ai, Xuanhui Wang, Sebastian Bruch, Nadav Golbandi, Mike Bendersky, **Marc Najork**. *Learning Groupwise Multivariate Scoring Functions Using Deep Neural Networks*. In Proceedings of the 5th ACM SIGIR International Conference on the Theory of Information Retrieval, pages 85–92, October 2019.
- C66 | Sebastian Bruch, Xuanhui Wang, Mike Bendersky, **Marc Najork**. *An Analysis of the Softmax Cross Entropy Loss for Learning-to-Rank with Binary Relevance*. In Proceedings of the 5th ACM SIGIR International Conference on the Theory of Information Retrieval, pages 75–78, October 2019.
- C65 | Rama Kumar Pasumarthi, Sebastian Bruch, Xuanhui Wang, Cheng Li, Mike Bendersky, **Marc Najork**, Jan Pfeifer, Nadav Golbandi, Rohan Anil, Stephan Wolf. *TF-Ranking: Scalable TensorFlow Library for Learning-to-Rank*. In Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pages 2970–2978, August 2019.
- C64 | Cheng Li, Mingyang Zhang, Mike Bendersky, Hongbo Deng, Don Metzler, **Marc Najork**. *Multi-view Embedding-based Synonyms for Personal Search*. In Proceedings of the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval, pages 575–584, July 2019.
- C63 | Sebastian Bruch, Masrour Zoghi, Mike Bendersky, **Marc Najork**. *Revisiting Approximate Metric Optimization in the Age of Deep Neural Networks*. In Proceedings of the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval, pages 1241–1244, July 2019.
- C62 | Aman Agarwal, Xuanhui Wang, Cheng Li, Mike Bendersky, **Marc Najork**. *Addressing Trust Bias for Unbiased Learning-to-Rank*. In Proceedings of the 2019 World Wide Web Conference, pages 4–14, May 2019.
- C61 | Shuguang Han, Bernhard Brodowsky, Przemek Gajda, Sergey Novikov, Mike Bendersky, **Marc Najork**, Robin Dua, Alexandrin Popescul. *Predictive Crawling for Commercial Web Content*. In Proceedings of the 2019 World Wide Web Conference, pages 627–637, May 2019.
- C60 | Furkan Kocayusufoglu, Ying Sheng, Nguyen Ha Vo, James B. Wendt, Qi Zhao, Sandeep Tata, **Marc Najork**. *RiSER: Learning Better Representations for Richly Structured Emails*. In Proceedings of the 2019 World Wide Web Conference, pages 886–895, May 2019.
- C59 | Jyun-Yu Jiang, Mingyang Zhang, Cheng Li, Mike Bendersky, Nadav Golbandi, **Marc Najork**. *Semantic Text Matching for Long-Form Documents*. In Proceedings of the 2019 World Wide Web Conference, pages 795–806, May 2019.
- C58 | Qingyao Ai, Xuanhui Wang, Nadav Golbandi, Michael Bendersky, **Marc Najork**. *Learning Groupwise Scoring Functions using Deep Neural Networks*. In WSDM 2019 Workshop on Deep Matching in Practical Applications, February 2019.
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<i>Dec 2021</i>	Special Section on Graph Technologies for User Modeling and Recommendation, Part 2 ACM Transactions on Information Systems, Volume 40, Issue 3, Article 42
<i>Sep 2021</i>	Special Section on Graph Technologies for User Modeling and Recommendation, Part 1 ACM Transactions on Information Systems, Volume 40, Issue 2, Article 21
<i>Feb 2008</i>	Special Section on Adversarial Issues in Web Search ACM Transactions on the Web, Volume 2, Issue 1, Article 1

CONFERENCE & WORKSHOP COMMITTEES

2027	WSDM (SPC) · ECIR (SPC)
2026	WSDM (associate chair) · ECIR (SPC) · SIGIR (program co-chair) · CIKM (SPC)
2025	WSDM (program co-chair) · SIGIR (SPC) · CIKM (SPC)
2024	WSDM (SPC) · ECIR (SPC) · SIGIR (SPC) · CIKM (industry day co-chair)
2023	WSDM (SPC) · ECIR (SPC) · SIGIR (SPC) · CIKM (SPC)
2022	WSDM (SPC) · SIGIR (SPC) · KDD (SPC) · CIKM (SPC)
2021	WWW (program co-chair) · SIGIR (SPC) · DESIRES (co-chair)
2020	WWW (SPC) · SIGIR (SPC) · BIG (co-chair)
2019	WSDM (SPC) · WWW (SPC) · SIGIR (short papers track co-chair) · KDD (SPC) · ICWE
2018	WSDM (SPC) · SIGIR · ASONAM · BIG
2017	WSDM (SPC) · ASONAM · BIG (co-chair)
2016	WSDM (SPC) · WWW
2015	WSDM (SPC) · CIKM (Industry track co-chair)
2014	–
2013	SIGIR (area chair)
2012	WSDM (SPC) · WWW · SIGIR (SPC)
2011	WSDM (tutorials co-chair) · WWW (PC & posters co-chair) · SIGIR (area chair)
2010	WSDM 2010 · WWW (Search track co-chair) · KDD (SPC)
2009	WSDM (SPC) · WWW (Industrial Practice & Experience track co-chair) · AIRWEB
2008	WSDM (general chair) · SIGIR · KDD · CIKM · AIRWEB · VLC
2007	WWW (tutorials & workshops co-chair) · AIRWEB · VLC
2006	WWW (Industrial Practice & Experience track co-chair) · SIGIR · AIRWEB (co-chair) · VLC · VL
2005	WWW · SIGIR · AIRWEB · VL
2004	WWW (program co-chair) · VLC
2003	WWW (Browsers and Tools track vice chair · IC · WAW · VLC
2002	WWW (Browsers and User Interfaces track deputy chair)
2001	WWW (Browsers and Tools track co-chair)
2000	WWW · VL
1999	WWW · VL (PC & video chair)
1998	VL
1997	–
1996	VL

STEERING COMMITTEES

2026–2029	ACM Web Conference (WWW) (vice-chair)
2017–2018	Conference on Design of Experimental Search and Information Retrieval Systems (DESIRES)
2017–2018	BigData Innovators Gathering (BIG)
2010–2018	ACM International Conference on Web Search and Data Mining (WSDM) (chair from 2014–2018)

OTHER SERVICE ACTIVITIES

2026–2028	Chair of the Seoul Test of Time Award Selection Committee
2025–2026	Chair of the ACM SIGIR Academy Selection Committee
2020–2026	Member of the ACM Publications Board
2024	Member of the IEEE Computer Society Fellow Evaluating Committee